FFHR – Farmer's Farm Health Report

Introduction

FFHR (Farmer's Farm Health Report) is an AI-powered, low-cost tool designed to help small and medium-scale farmers understand their soil health and make informed farming decisions. The system uses a combination of simple physical tests, location data, and AI to generate easy-to-understand farm reports.

Detailed Workflow

1. **Input Collection**
    - The farmer performs a pH test using a strip.
    - A cow dung patch test is done and an image is captured.
2. **Data Gathering**
    - GPS coordinates and date/time are collected via mobile.
    - APIs are used to fetch local soil and weather data (e.g., SoilGrids, NASA POWER).
3. **AI Analysis**
    - AI analyzes the cow dung patch image.
    - Combines image data with pH, location, weather, and soil data.
4. **Report Generation**
    - Generates a digital Farm Health Report.
    - Includes soil condition, recommended crops, fertilizer guidance, and sustainability suggestions.
5. **Farmer Access**
    - Report can be accessed on mobile or printed for offline use.

Benefits

- Affordable soil testing for remote/rural areas
- Reduces chemical misuse
- Encourages climate-resilient farming
- No lab or sensor dependency

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